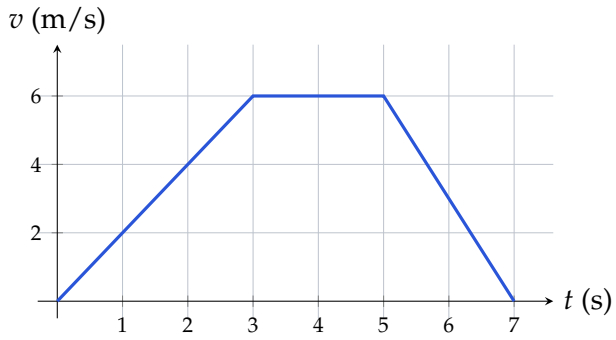
AP PHYSICS 1

Unit 1 • Lesson 1.3 • Notes (filled in)

Topic 1.3 • Representing Motion

Notes (filled in)

Objective: read and draw position, velocity and acceleration graphs, and move between them.

Position vs. time	On an x–t graph, the slope is the velocity. Straight line: constant velocity. Curved line: the velocity changes (acceleration). Horizontal line: the object is at rest.
Velocity vs. time	On a v–t graph: the slope is the acceleration; the area between the graph and the t-axis is the displacement. Area below the axis counts as negative displacement. Where the graph crosses the axis, the object turns around.
Acceleration vs. time	On an a–t graph, the area is the change in velocity Δv.
Example: a v–t graph	 Accelerations: 0–3 s: $2\,\mathrm{m/s^2}$ 3–5 s: 0 5–7 s: $-3\,\mathrm{m/s^2}$ Displacement = total area = $9 + 12 + 6 = 27\,\mathrm{m}$ Average velocity over 7 s: $27/7 \approx 3.9\,\mathrm{m/s}$
Motion diagrams	Dots show the position at equal time intervals. ● ● ● ● ● ● equal spacing: constant velocity ●● ● ● ● ● spacing grows: speeding up
Summary	Slope of x–t = velocity; slope of v–t = acceleration; area under v–t = displacement; area under a–t = change in velocity. A v–t graph crossing the axis means a change of direction.